

Task 1: Print even numbers using while loops, do-while loops and array.

Task Name: Write a java program that takes starting and ending numbers from user prints all even numbers in that range using array.

Objective:

- To learn input-taking processes in Java
- To understand while loop usage
- To store data in an array
- To identify even numbers from a range

Problem Analysis:

The program will take two numbers from the user as starting and ending values. Then the program will check each number. If the number is divisible by 2, it will be considered an even number. These even numbers will be saved into an array and later printed.

Algorithm:

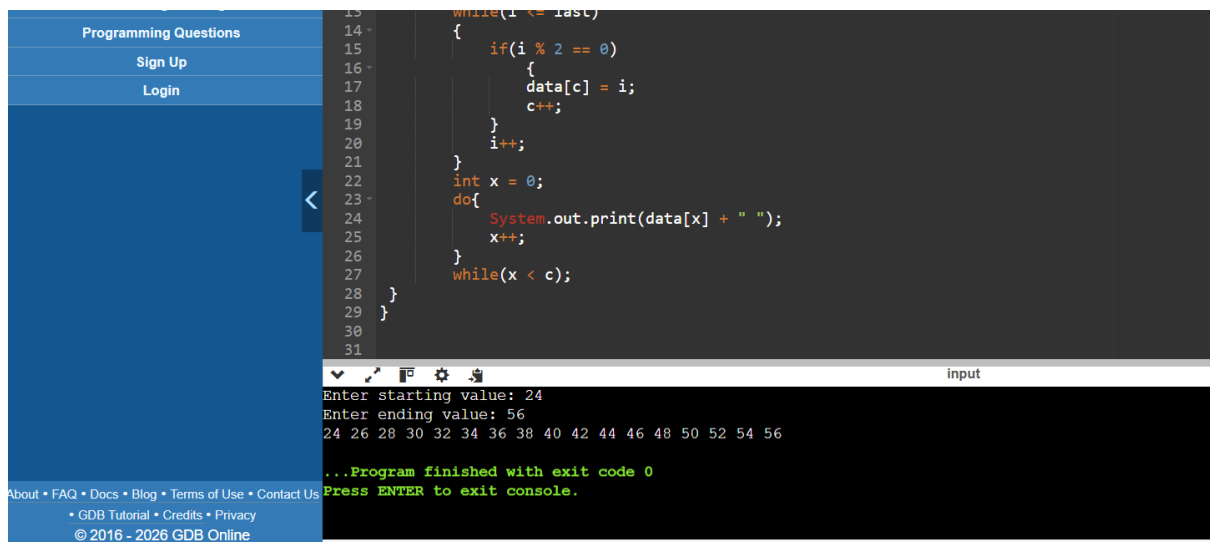
Step 1: Start the program.
Step 2: Input lower limit and upper limit.
Step 3: Declare an array to store even numbers.
Step 4: Initialize loop variable.
Step 5: Use while loop to check numbers.
Step 6: Store even numbers into array.
Step 7: Use do-while loop to display numbers.
Step 8: Stop.

Source Code:

```
import java.util.Scanner;
public class EvenTask {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int first, last;
        System.out.print("Enter starting value: ");
        first = sc.nextInt();
        System.out.print("Enter ending value: ");
        last = sc.nextInt();
        int data[] = new int[100];
        int i = first;
        int c = 0;
        while(i <= last)
        {
            if(i % 2 == 0)
```

```
        {
            data[c] = i;
            c++;
        }
        i++;
    }
    int x = 0;
    do{
        System.out.print(data[x] + " ");
        x++;
    }
    while(x < c);
}
```

Output:



```
13 while(i <= last)
14 {
15     if(i % 2 == 0)
16     {
17         data[c] = i;
18         c++;
19     }
20     i++;
21 }
22 int x = 0;
23 do{
24     System.out.print(data[x] + " ");
25     x++;
26 } while(x < c);
27 }
28 }
29 }
30 }
31 }
```

input

Enter starting value: 24
Enter ending value: 56
24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56
...Program finished with exit code 0
Press ENTER to exit console.

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Discussion & Conclusion:

This program is designed to print all even numbers within a user-defined range using an array in Java. The starting and ending numbers are taken as input from the user, and the values are stored in an array. The program then checks each number using the modulus operator (%) to determine whether it is even. If a number is divisible by 2, it is displayed as an even number. This experiment helps in understanding the practical use of arrays, loops, conditional statements, and user input handling in Java programming. The experiment was successfully completed, and the program correctly displayed all even numbers within the given range. It improved understanding of array operations and logical conditions in Java. The program is simple, efficient, and useful for performing mathematical operations through programming.

Task 2: Store and print prime numbers in an array.

Task Name:

Write a Java that takes starting and ending numbers from the user and stores only prime numbers in an array, and prints them.

Objective:

- To understand prime number logic
- To practice nested loop
- To learn array storing method
- To improve Java programming basics
- To filter numbers based on condition

Problem Analysis:

The program will take two numbers as range. Then each number will be checked whether it is prime or not. If the number has exactly two divisors then it is prime. Prime numbers will be stored in an array and displayed at the end.

Algorithm:

Step 1: Start
Step 2: Input start number
Step 3: Input end number
Step 4: Take an array
Step 5: Check each number
Step 6: Count divisors
Step 7: If divisor = 2 store number
Step 8: Print stored values
Step 9: End

Source Code:

```
import java.util.Scanner;
public class PrimeTask {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int start, end;
        System.out.print("Enter start: ");
        start = sc.nextInt();
        System.out.print("Enter end: ");
        end = sc.nextInt();
        int arr[] = new int[100];
        int count = 0;
        for(int i = start; i <= end; i++)
```

```
        if(factor == 2)
        {
            arr[count] = i;
            count++;
        }
    }
    System.out.println("Prime numbers
list:");
    for(int i = 0; i < count; i++)
    {
        System.out.print(arr[i] + " ");
    }
}
```

<pre> { int fact = 0; for(int j = 1; j <= i; j++) { if(i % j == 0) { fact++; } } } </pre>	<pre> } </pre>
--	----------------

Output:

```

code, compile, run, debug, share.
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2 public class PrimeNumber {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int start, end;
6         System.out.print("Enter start: ");
7         start = sc.nextInt();
8         System.out.print("Enter end: ");
9         end = sc.nextInt();
10        int arr[] = new int[100];
11        int count = 0;
12        for(int i = start; i <= end; i++)
13        {
14            int fact = 0;
15            for(int j = 1; j <= i; j++)
16            {
17                if(i % j == 0)
18                {
19                    fact++;
20                }
21            }
22            if(fact == 2)
23            {
24                arr[count] = i;
25                count++;
26            }
27        }
28        System.out.print("Prime numbers list:");
29        for(int i = 0; i < count; i++)
30        {
31            System.out.print(arr[i] + " ");
32        }
33        System.out.println();
34    }
35}

Enter start: 3
Enter end: 20
Prime numbers list:
3 5 7 11 13 17 19

...Program finished with exit code 0
Press ENTER to exit console.

```

Discussion & Conclusion: This program is designed to find and store prime numbers within a user-defined range using an array in Java. The starting and ending values are taken as input from the user, and each number in the range is checked to determine whether it is prime or not. A number is considered prime if it is divisible only by 1 and itself. The prime numbers are stored in an array and then printed as output. This experiment helps in understanding the use of arrays, loops, conditional statements, and prime number logic in Java programming. It also improves problem-solving skills and logical thinking. The program was successfully executed and correctly stored and displayed all prime numbers within the given range. Through this experiment, the concepts of arrays, loops, and conditional checking in Java became clearer. The program is efficient and useful for mathematical computations involving prime numbers.